

Applied Linear Algebra (EE 635): Problem Set 3

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August 15, 2025

Questions

1. Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for $\mathbb{R}^n$. Given a coordinate vector $[\mathbf{x}]_B$ of a vector $\mathbf{x}$ relative to B , find the vector $\mathbf{x}$.

- (a) Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be a basis for $\mathbb{R}^3$, where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}.$$

If

$$[\mathbf{x}]_B = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix},$$

find $\mathbf{x}$ in the standard basis of $\mathbb{R}^3$.

- (b) Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5\}$ be a basis for $\mathbb{R}^5$, where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b}_4 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{b}_5 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}.$$

If

$$[\mathbf{x}]_B = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \\ 4 \end{bmatrix},$$

find $\mathbf{x}$ in the standard basis of $\mathbb{R}^5$.

2. Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for $\mathbb{R}^n$. Given a vector $\mathbf{x}$ in the standard basis of $\mathbb{R}^n$, find its coordinate vector $[\mathbf{x}]_B$ relative to the basis B .

- (a) Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be a basis for $\mathbb{R}^3$, where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}.$$

If

$$\mathbf{x} = \begin{bmatrix} 5 \\ 4 \\ 10 \end{bmatrix},$$

determine $[\mathbf{x}]_B$.

- (b) Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5\}$ be a basis for $\mathbb{R}^5$, where

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b}_4 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{b}_5 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}.$$

If

$$\mathbf{x} = \begin{bmatrix} 7 \\ 2 \\ 4 \\ 5 \\ 6 \end{bmatrix},$$

determine $[\mathbf{x}]_B$.

3. Let A be an $n \times n$ matrix and suppose A is invertible. Show that the following statements hold:

- (a) A is row equivalent to the identity matrix I_n .
- (b) $\dim(\text{Col}(A)) = n$.
- (c) $\ker(A) = \{\mathbf{0}\}$.
- (d) $\dim(\text{Row}(A)) = n$.

(Prove each item directly from the assumption that A is invertible.)

4. (a) Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ -5 \\ 8 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 8 \\ 2 \\ -9 \end{bmatrix}.$$

Determine if $\mathbf{w}$ is in the subspace of $\mathbb{R}^3$ generated by $\{\mathbf{v}_1, \mathbf{v}_2\}$.

(b) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ -7 \\ 9 \\ 7 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 5 \\ -8 \\ 6 \\ 5 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} -4 \\ 10 \\ -7 \\ -5 \end{bmatrix}.$$

Determine if $\mathbf{u}$ is in the subspace of $\mathbb{R}^4$ generated by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

(c) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Determine if $\mathbf{w}$ is in the subspace of $\mathbb{R}^3$ generated by $\{\mathbf{v}_1, \mathbf{v}_2\}$.

5. Given two subspaces W_1 and W_2 , find a basis for the intersection $W_1 \cap W_2$ and the sum $W_1 + W_2$.

(a) In $\mathbb{R}^3$, let

$$W_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}, \quad W_2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(b) In $\mathbb{R}^4$, let

$$W_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \end{bmatrix} \right\}, \quad W_2 = \text{span} \left\{ \begin{bmatrix} 0 \\ -1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \\ -2 \end{bmatrix} \right\}.$$

6. Let A be an $m \times n$ matrix over $\mathbb{R}^n$.

(a) Prove that the *kernel* of A , defined as

$$\ker(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\},$$

is a subspace of $\mathbb{R}^n$.

(b) Prove that the *image* of A , defined as

$$\text{im}(A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\},$$

is a subspace of $\mathbb{R}^m$.